USER MANUAL · POST-PROCESSING

Looking at the results: the 3D Results Viewer

How to open the finished demo room fire in 3D, play it like a film, cut it open, read values by clicking, plot the output files and save pictures and an MP4 animation, or record your own walk-through in real time. Every screenshot is the real window, working on the real run of the demo case.

> THE DEMO CASE
>
> **A gas burner is lit in the middle of a small room, and the run covers the first 20 seconds.** The room is 4.8 × 3.3 m and 2.4 m high with a door and a window. The burner grows to about 360 kW. FDS wrote two temperature slices, 3D smoke and flame data (655 frames), Plot3D files and two CSV files, in `examples/example-1/gui_run/` .
>
> Start the viewer from the main window with **Results > View Results (3D)...**, then **File > Open Results (.smv)...** and pick `gui_run/example_1.smv` . The ready-made setup `example_1.pyfdsv` opens the same picture in one step (File > Open Results Visualization...).

`Application: PyFDS desktop  Solver: FDS 6.10.1  Case: example_1  18 Oct 2026`

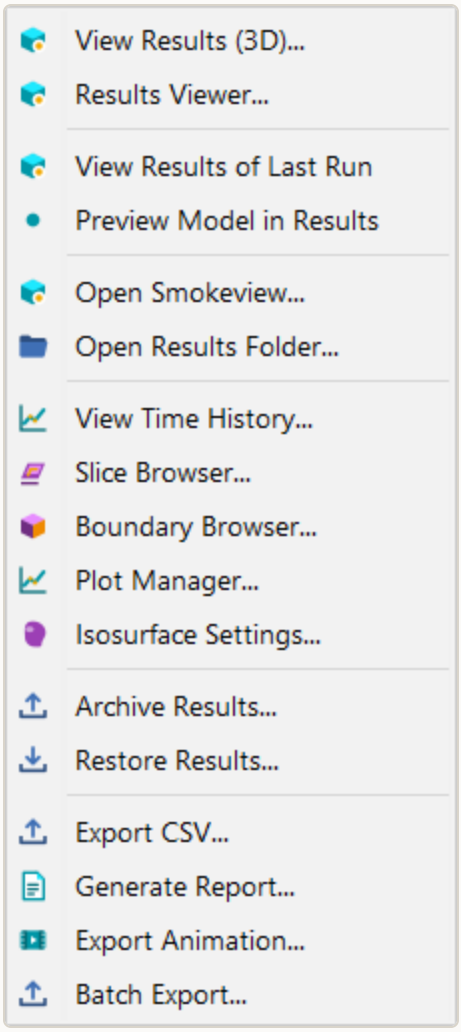
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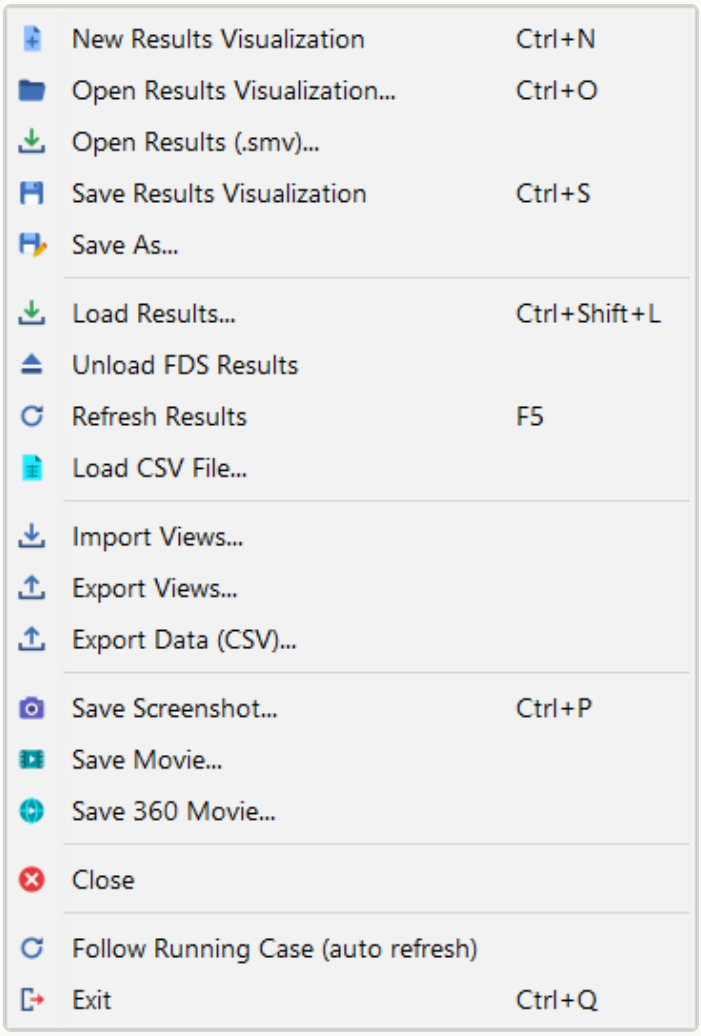
1 · Open the viewer

MENU Results > View Results (3D)...

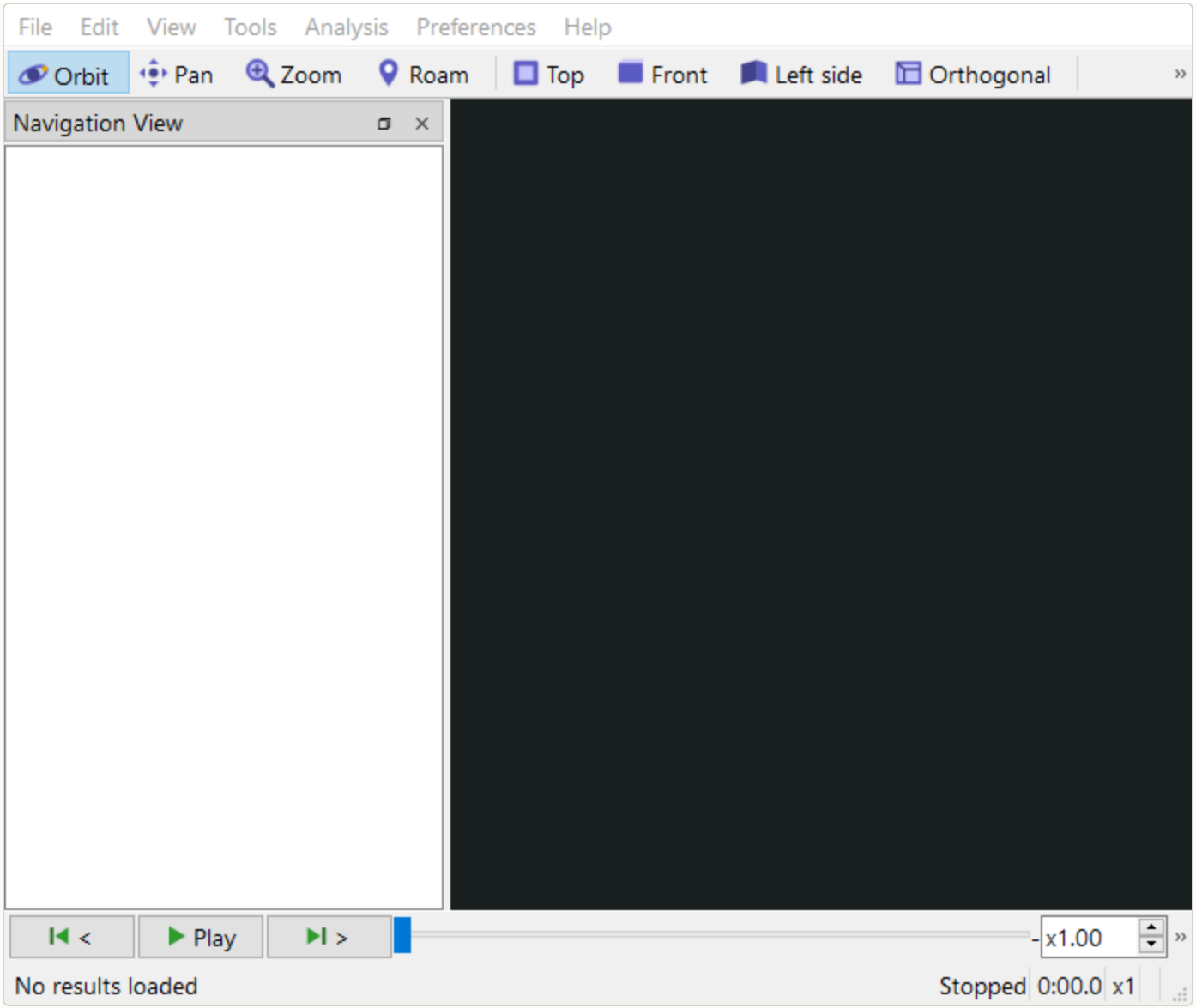
The entry sits at the top of the Results menu of the main window.



Results > View Results (3D)...



The viewer's own File menu: open or load results, load a CSV file, save screenshots and movies.



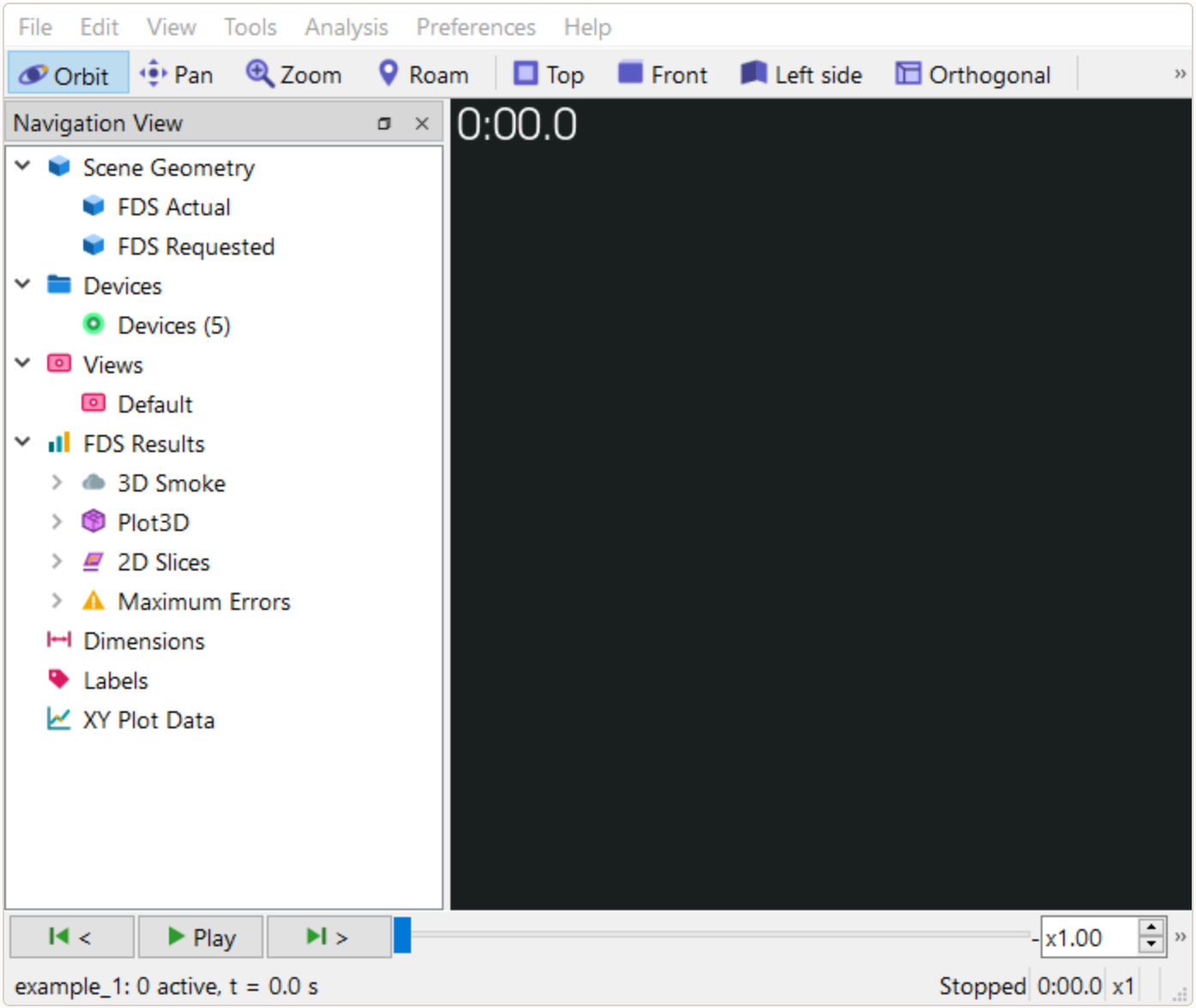
The viewer before anything is loaded. **Left:** the Navigation View (a tree of everything that can be shown). **Centre:** the 3D scene. **Bottom:** the time bar with Play, a slider, speed and Loop. **Top:** menus and a toolbar with the camera tools and rendering mode.

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2 · Load the results of the run

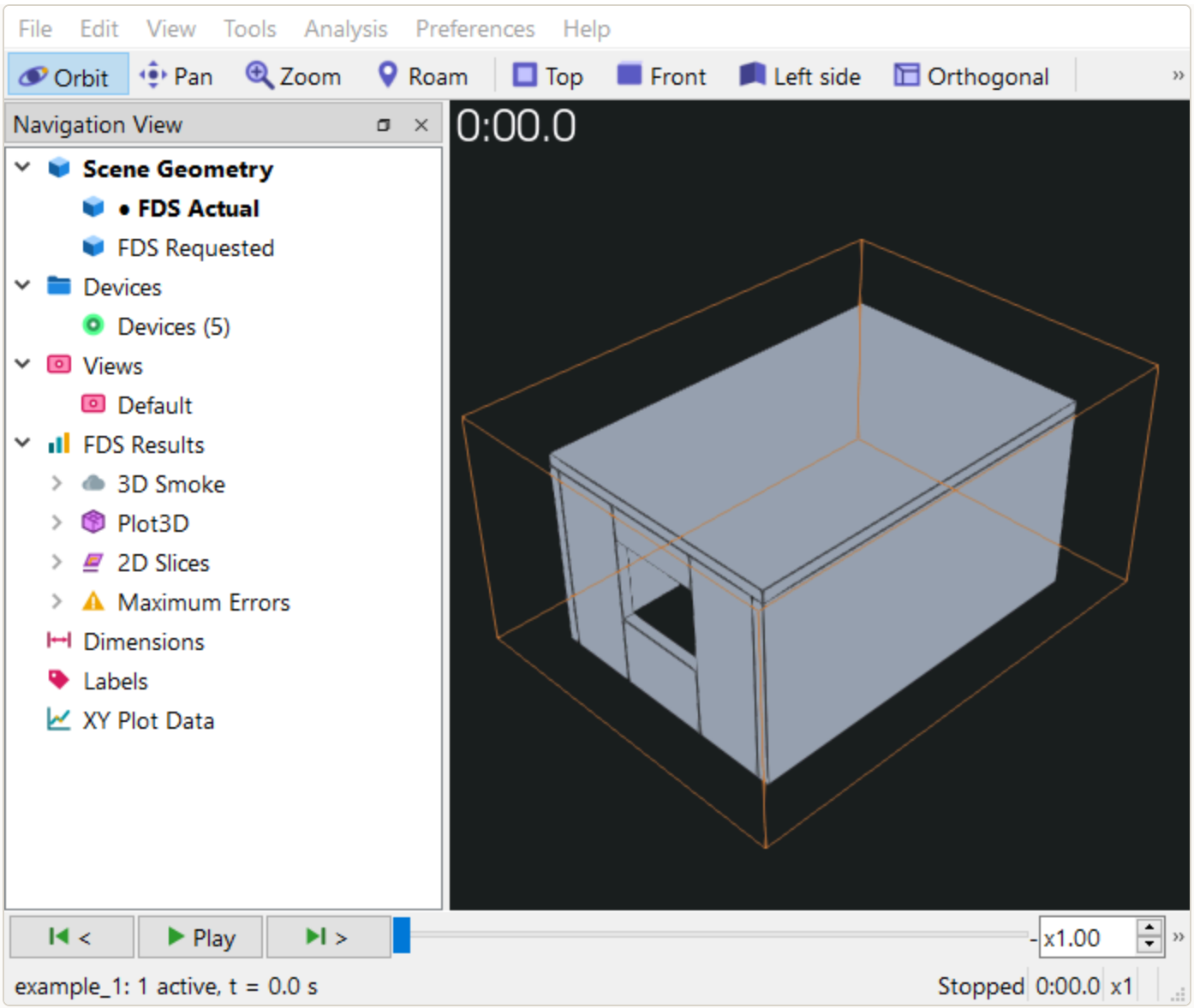
`MENU` File > Open Results (.smv)... then pick `example_1.smv` in the run folder

The `.smv` file is the index FDS writes next to its data. The viewer reads it and builds the tree: scene geometry, **3D Smoke** (smoke and flame), **Plot3D** (the whole room at chosen times), **2D Slices** (the slices requested in the model), and the solver's **Maximum Errors**. Nothing is drawn until you switch an item on. The case here has two slices, three kinds of smoke data and four Plot3D quantities.



Results loaded. The tree is full; the scene is empty until something is switched on.

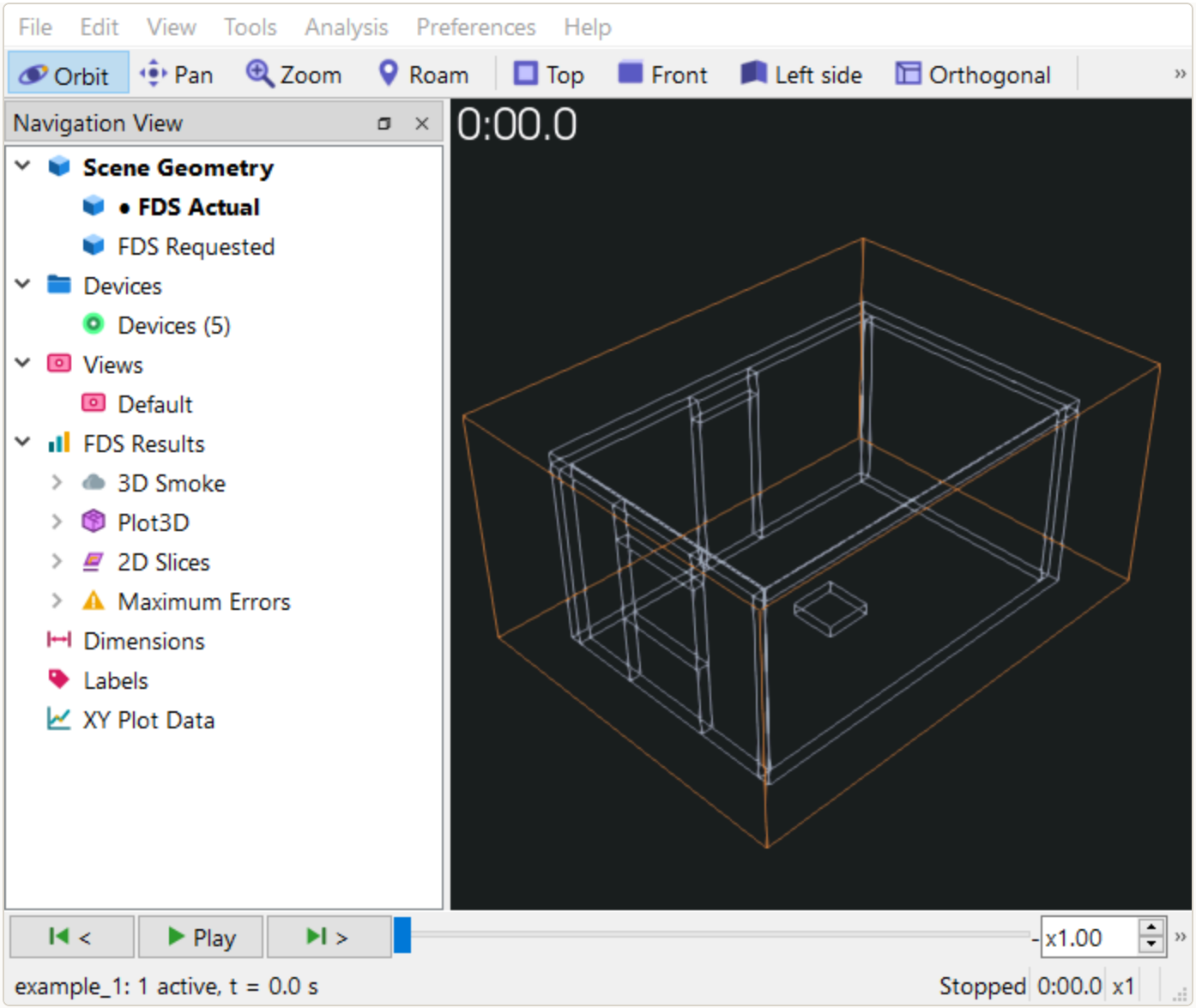
Double-click an item in the tree to switch it on or off (active items are shown in bold). Switch on **FDS Actual** under Scene Geometry to see what FDS really built: the walls, door, window and burner, inside the outline of the computational mesh.



FDS Actual geometry. The orange outline is the open boundary of the domain; the grey boxes are the walls, ceiling, door and window opening.

Rendering mode

The box in the toolbar chooses how solids are drawn (Solid, Solid outlines, Realistic, Realistic outlines, Wireframe). The next figures use **Wireframe** so the inside of the room stays visible.



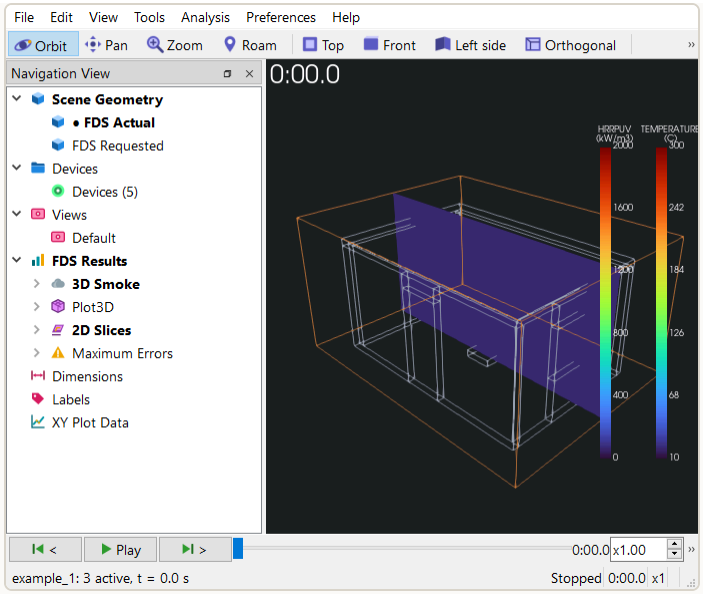
Wireframe: walls drawn as edges only.

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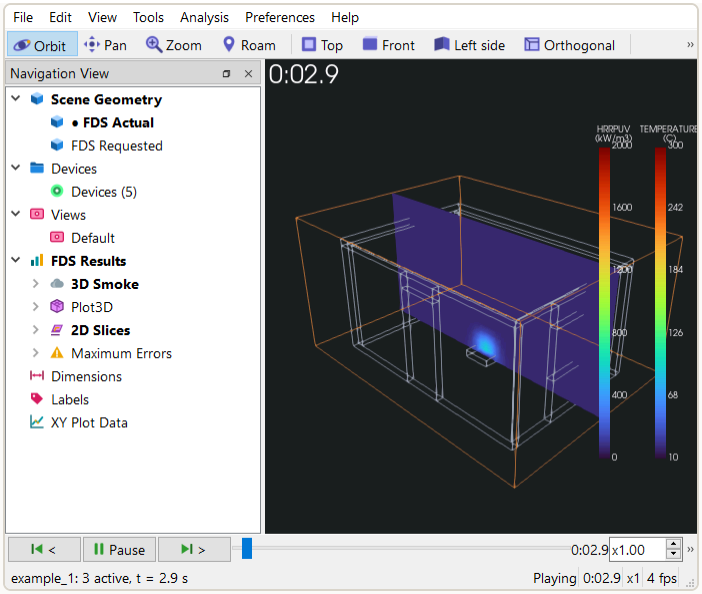
3 · Play the run

`TIME BAR` Play / Pause · < and > (one frame) · slider · speed (×1.00, ÷2, ×2) · Loop

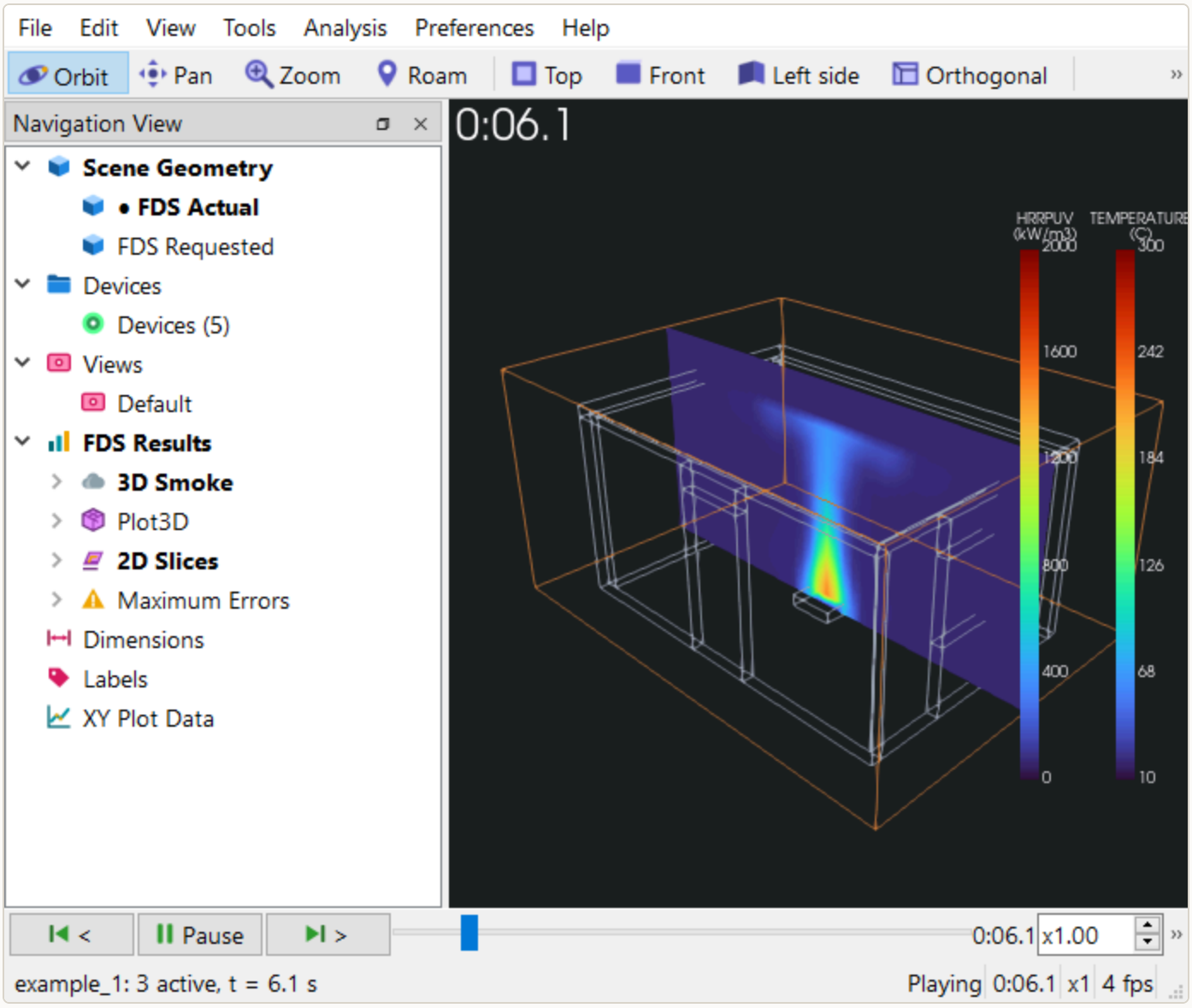
Switch on the walls (**FDS Actual**), a slice and the flame, then press **Play** at the bottom left. The clock at the top left of the scene and the slider move together, the colours of the slice follow the fire, and the status bar shows **Playing**, the speed and the frame rate. Press **Pause** to stop on any frame; **<** and **>** then step one frame at a time. Change the speed with the **×** box or the **÷2** and **×2** buttons, and tick **Loop** to start again at the end.



Ready: walls (wireframe), the slice and the flame are on, the clock is at 0.



Playing, a few seconds in: the Pause button, the moving clock and the frame rate in the status bar.



A few seconds later (t = 6.4 s): the plume has grown. Playback follows the real clock: if the computer cannot draw every frame in time, some frames are skipped, so the film does not run in slow motion.

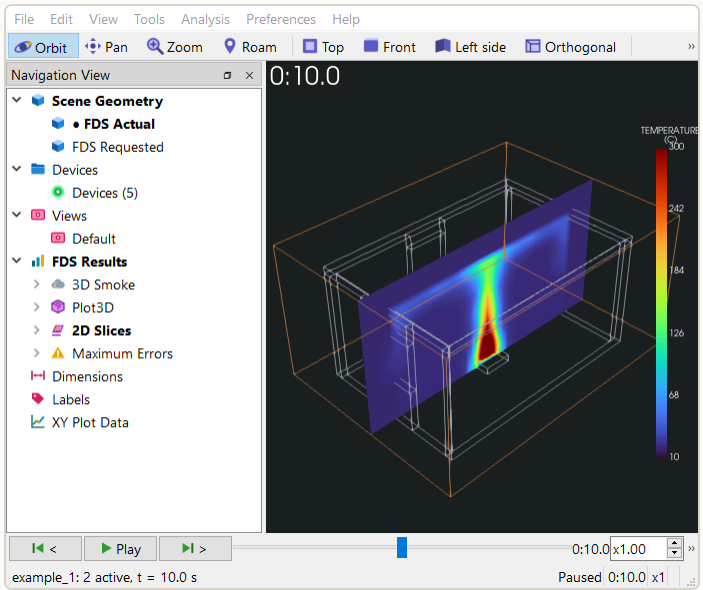
> **Measured on this case**
>
> Walls + slice + flame play at about **3 to 4 frames per second** on an ordinary laptop with an Intel HD 4600 graphics chip, and 1 s of film is 1 s of fire at ×1.00. Use ×2 or ×4 to get through the 20 s faster. The run has 655 flame frames; drawing every one at full size is not needed to follow the fire.

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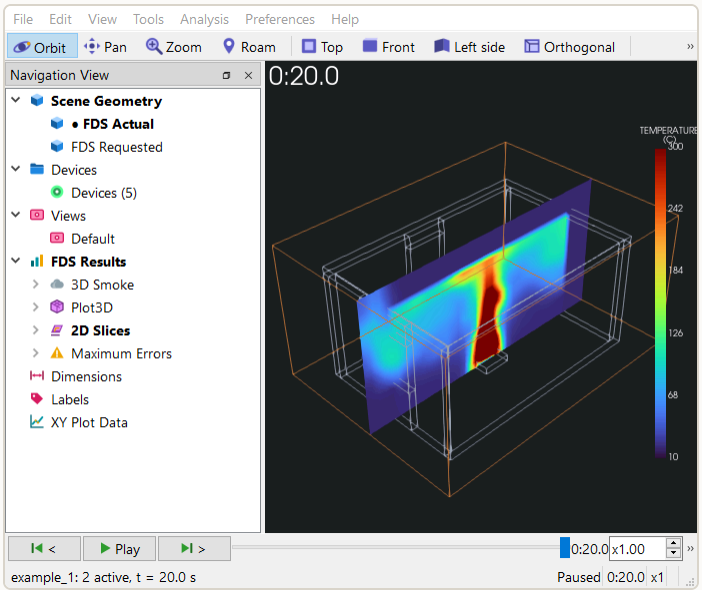
4 · Slices and the time bar

`TREE` 2D Slices > TEMPERATURE > SLICE_T_Y > Mesh 1 (double-click) · `TIME BAR` slider, Play, speed, Loop

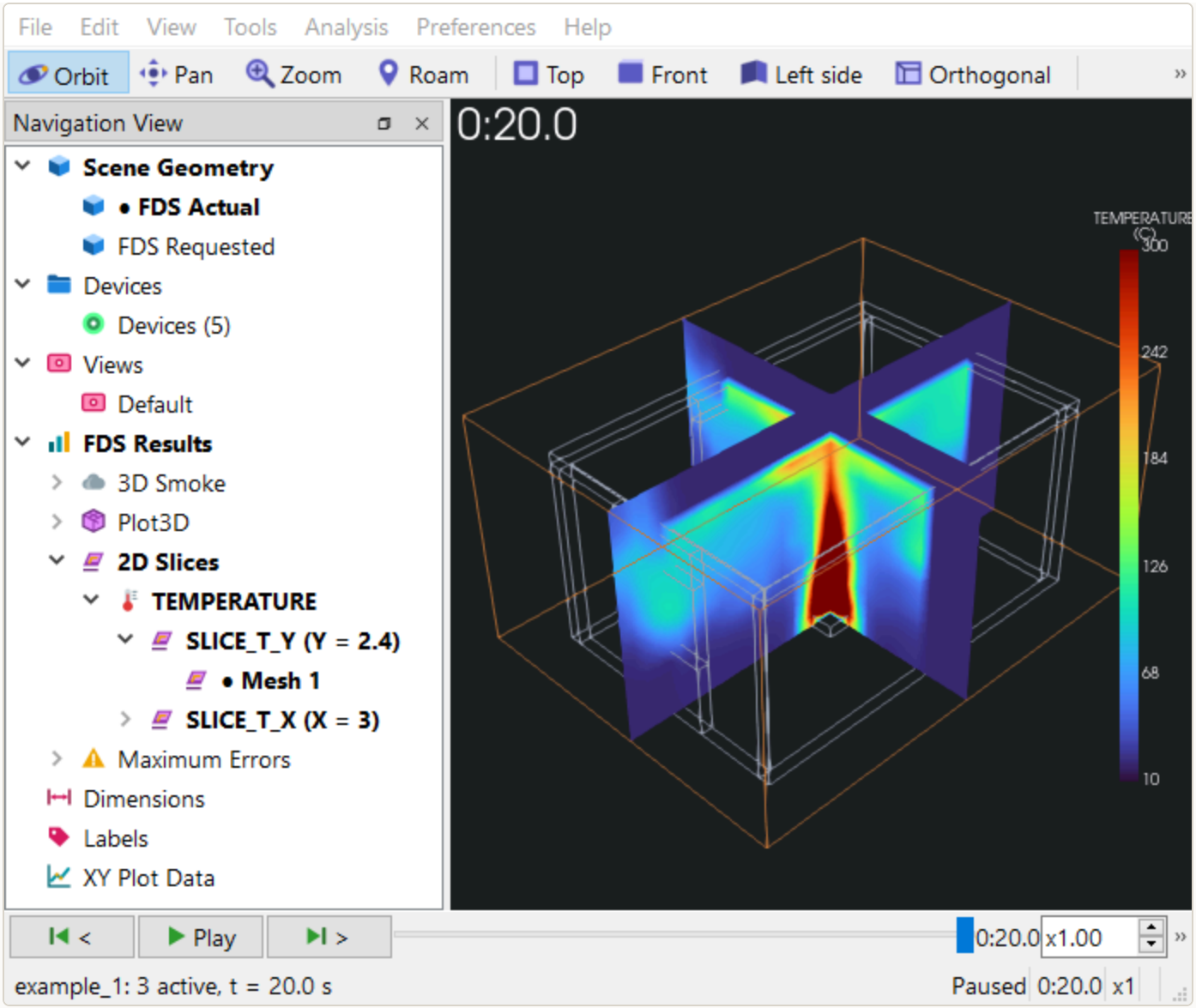
A slice is a flat cut through the room that shows one quantity in colour. Switching it on adds a **colour bar** at the right and a clock at the top left. Drag the slider to any time, step with **<** and **>**, or press **Play**. The speed box (×1.00) and the ÷2 and ×2 buttons change the playback speed; **Loop** repeats.



t = 10 s: the plume rises from the burner and starts to spread under the ceiling.



t = 20 s: a hot layer under the whole ceiling.



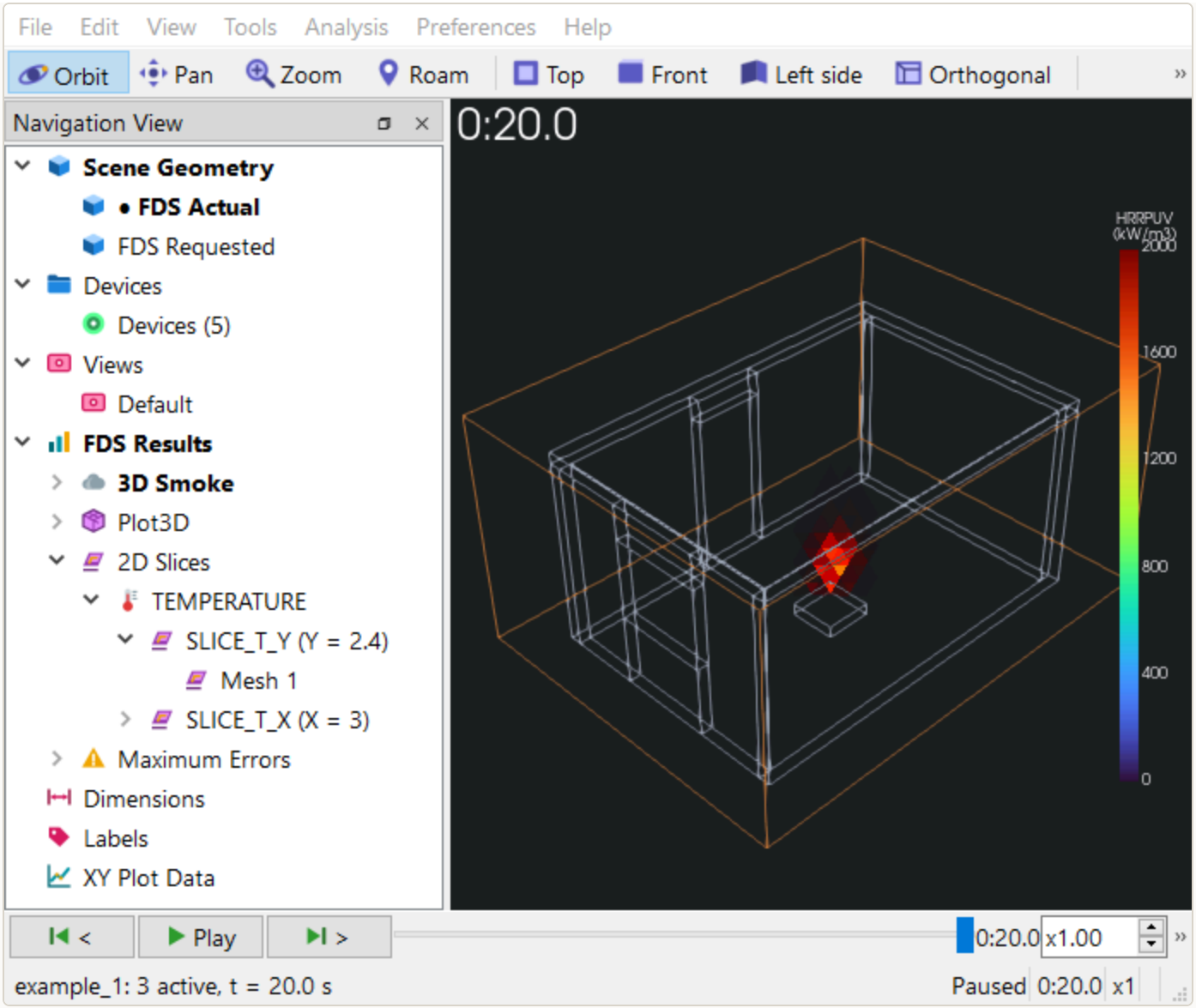
Both slices on at once (planes y = 2.4 m and x = 3.0 m). They cross exactly in the plume.

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5 · 3D smoke and flame

TREE 3D Smoke > HRRPUV > Mesh 1 (flame) and SOOT DENSITY > Mesh 1 (smoke)

3D Smoke shows the fire itself as a volume, not a flat cut. **HRRPUV** (heat release rate per unit volume) is the flame: the brighter, the more heat is released in that spot. **SOOT DENSITY** is the smoke. In this short run with a clean-burning gas fire there is almost no soot, so the flame is the part to look at. Look at it from different sides by dragging with the mouse (Orbit), or use the Top, Front and Left side buttons.



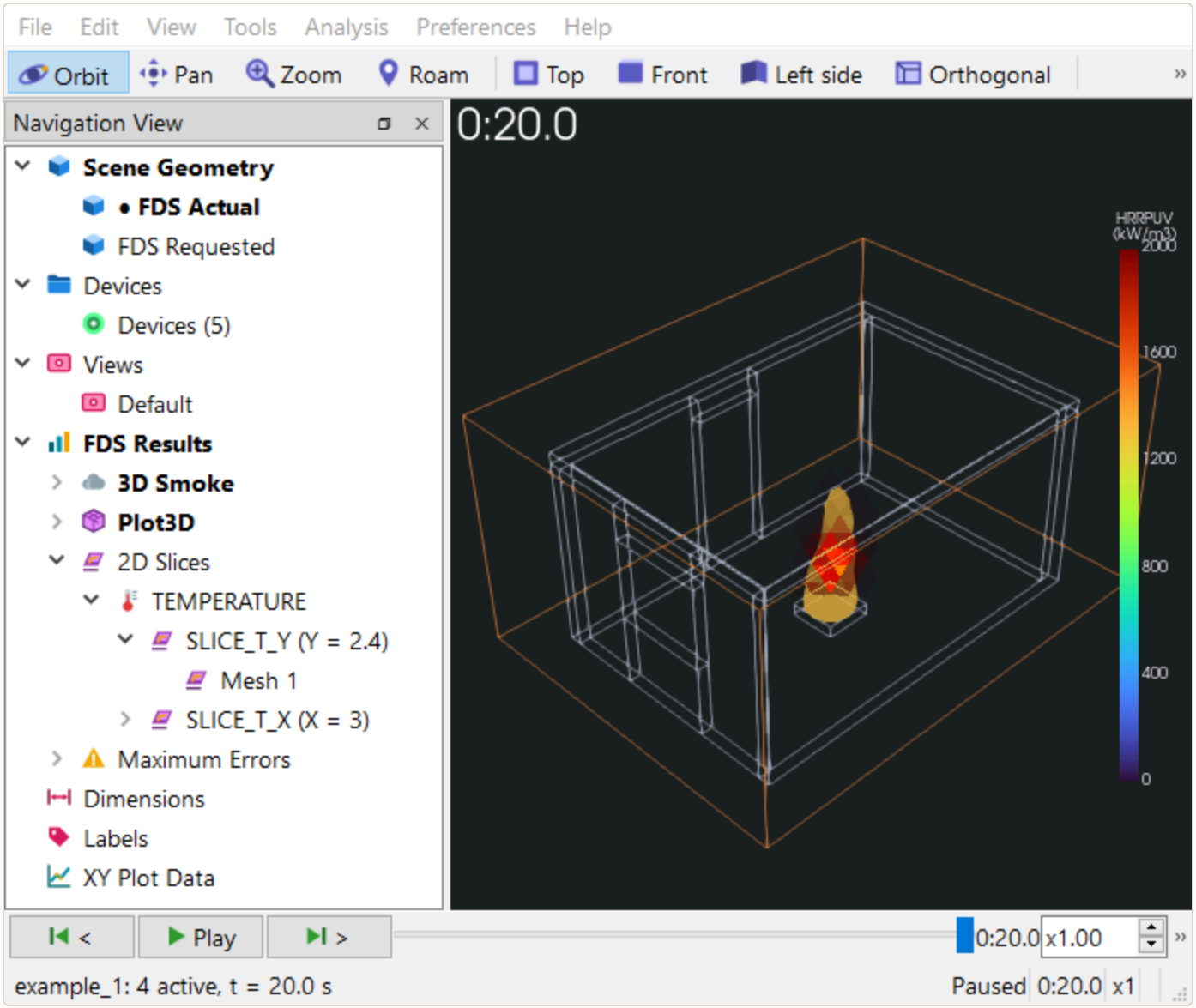
The flame at t = 20 s, drawn as a volume above the burner.

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6 · Isosurfaces from Plot3D

MENU Tools > Data3D: Add Isosurface… · choose the quantity (Plot3D > TEMPERATURE), enter the value (300), then switch the new item on in the tree

Plot3D files hold the whole room at a few times. An **isosurface** is the skin of all points that have exactly one value, here 300 °C. It shows the shape of the hot plume as a solid body.



The 300 °C surface (yellow) at t = 20 s: a tall column over the burner that has not yet reached the ceiling.

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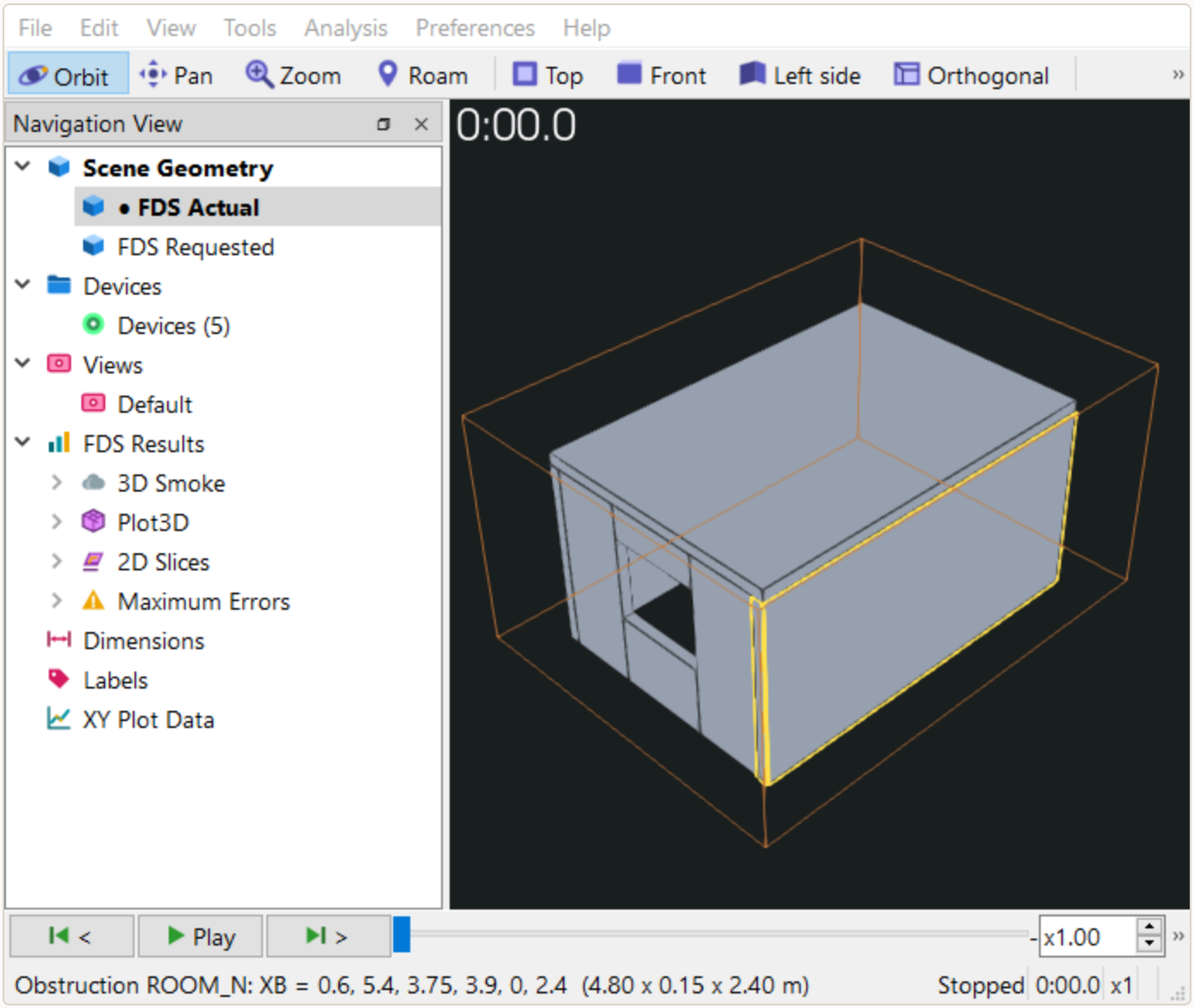
7 · Select objects by clicking

`MOUSE` click an object in the 3D scene · `ESC` or click empty space to clear

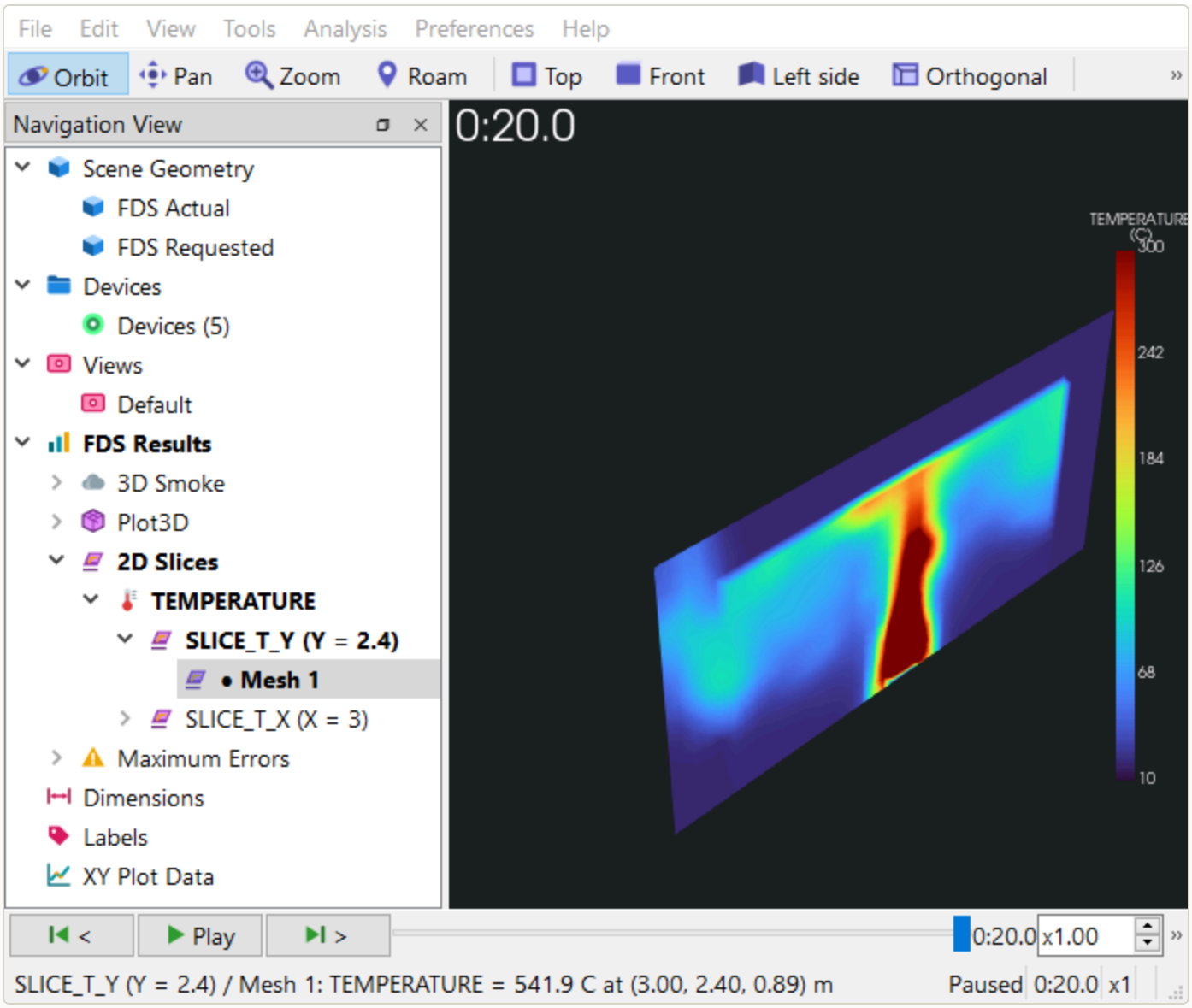
Click any object in the scene to find out what it is. A click is a press and release without moving the mouse; if you drag, the view turns instead. The object you hit is **outlined in yellow**, its row in the Navigation View is selected, and the **status bar** at the bottom describes it:

- **A wall, vent or other solid:** its name, its position (XB) and its size. In the figure below, the click on the north wall reports `Obstruction ROOM_N: XB = 0.6, 5.4, 3.75, 3.9, 0, 2.4 (4.80 × 0.15 × 2.40 m)`.

- **A slice or 3D data:** the value under the pointer. Clicking the temperature slice at the base of the plume reports `TEMPERATURE = 541.9 C at (3.01, 2.40, 0.89) m`.
- **An isosurface:** its level.



A click on the north wall: yellow outline, the tree row is selected, and the status bar gives the name and size.



A click on the slice at 0.89 m above the floor: the status bar reads 541.9 °C at that point (t = 20 s).

Good to know

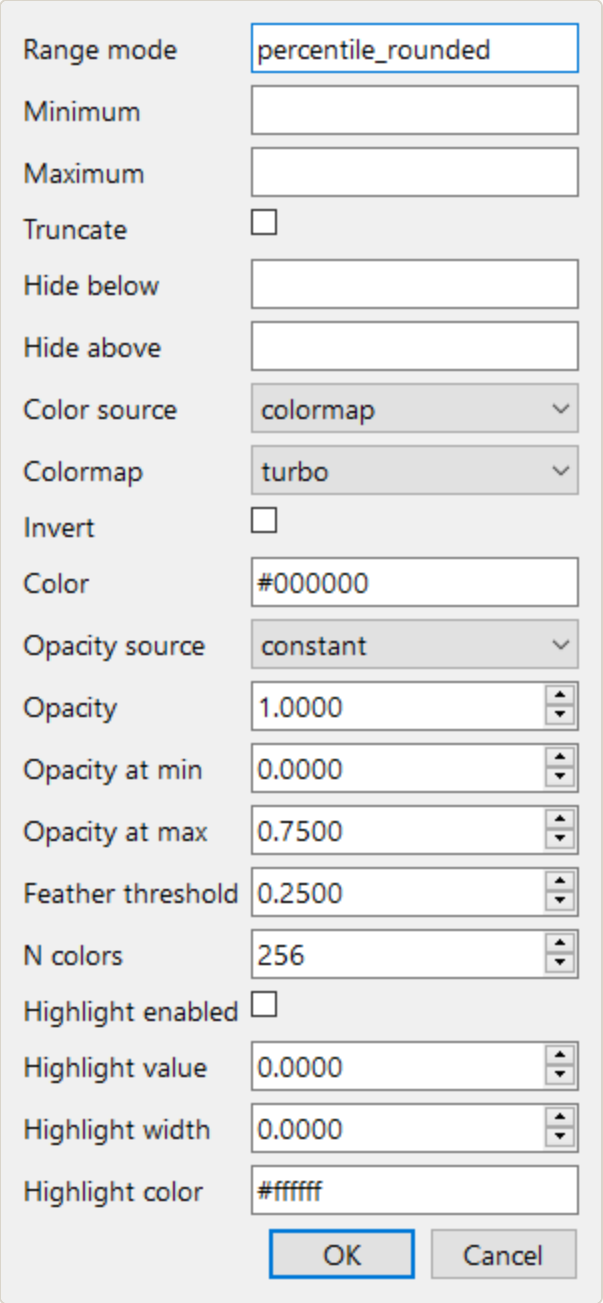
The open sides of the domain (the orange outline) are not clickable, so they never block the objects behind them. A click selects one object at a time; the 3D flame and smoke volumes cannot be selected by clicking.

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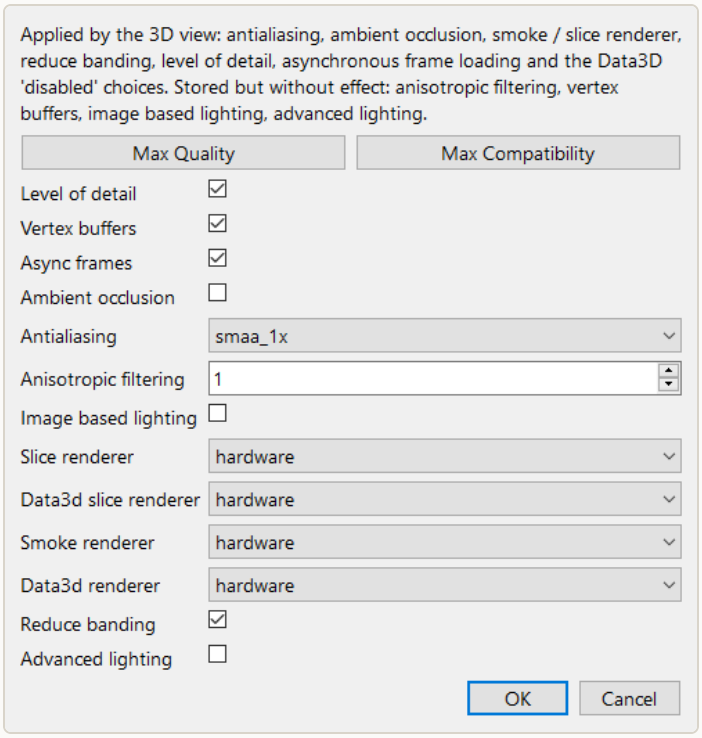
8 · Colour bar and rendering quality

MENU Preferences > Colorbar... (pick the quantity first) · Preferences > Rendering...

The **Colorbar** dialog sets the range of the colours (minimum, maximum, or automatic), the colour map (turbo, viridis, jet, ...), whether values below or above a limit are hidden, and how see-through the data is. The **Rendering** dialog controls the picture quality. **Max Quality** switches on everything the graphics card can do and **Max Compatibility** picks the safe options for an older machine; the note at the top says which options are applied.



Colorbar settings for TEMPERATURE.

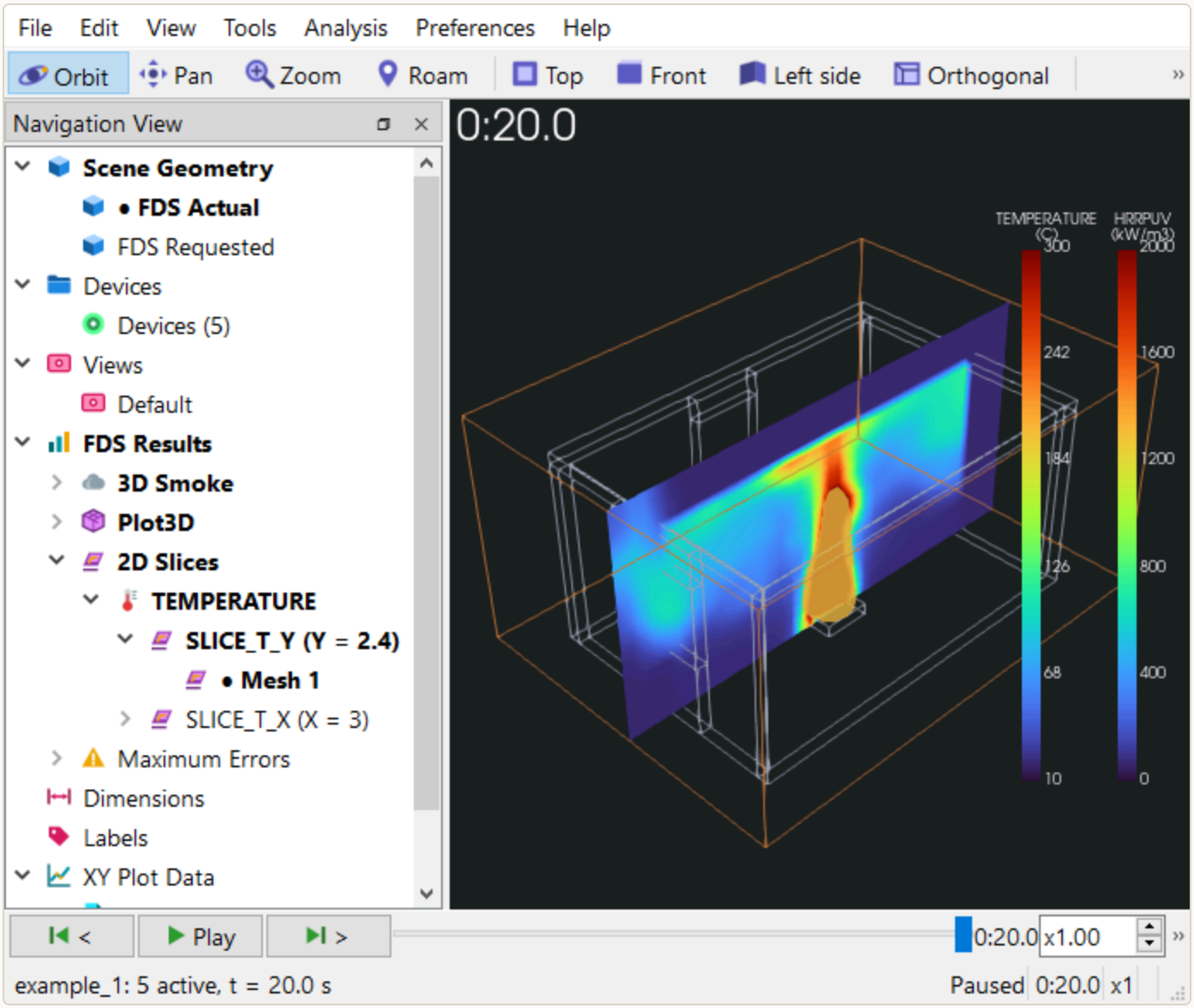


Rendering preferences with the two quality presets.

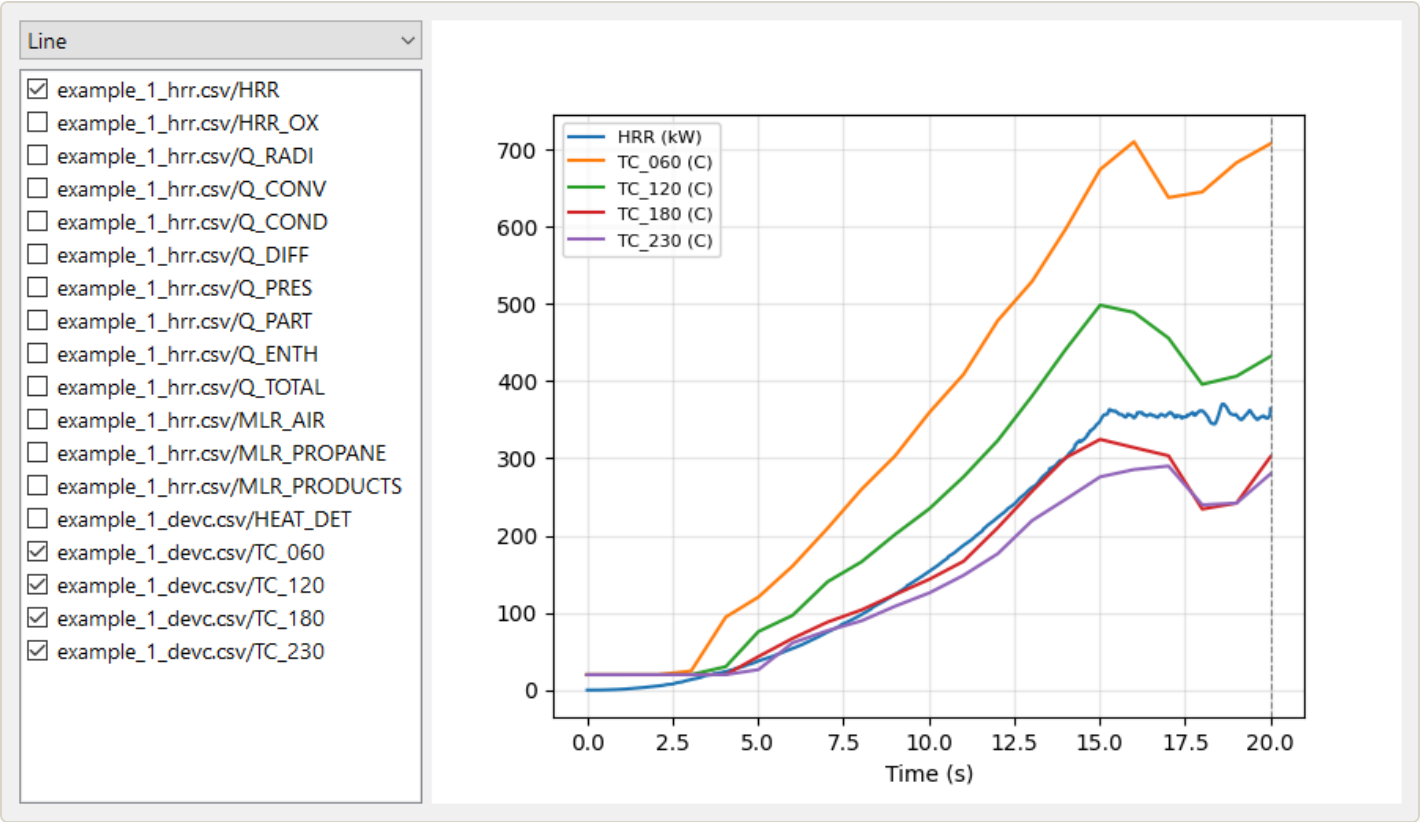
9 · Plots from the output files

`MENU` File > Load CSV File... (twice: `example_1_hrr.csv`, `example_1_devc.csv`) · Tools > Time History Plot...

The heat release file and the device file written by FDS can be loaded next to the 3D data. They appear under **XY Plot Data**. The plot window lists every column; tick the ones you want. A dashed line marks the time that the 3D scene is showing, so the curves and the picture can be read together. The plot type can be Line, Stem or Step, and **F11** saves the plot as an image.



Both CSV files loaded and listed in the tree.



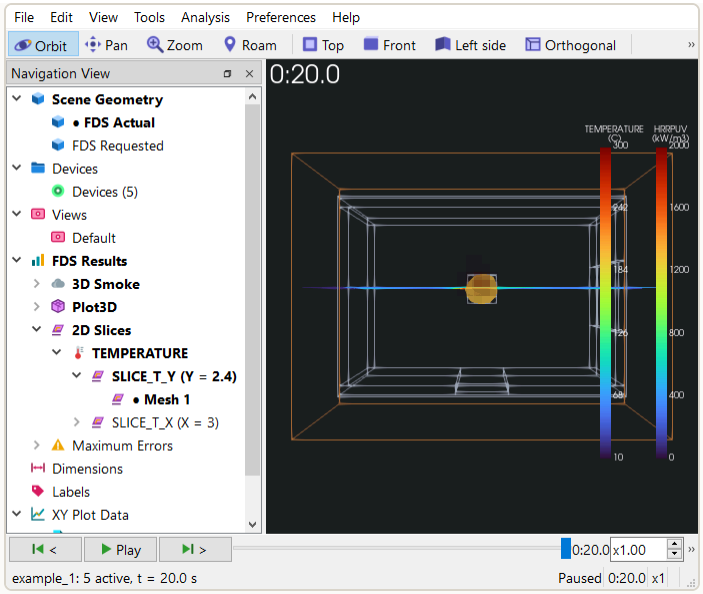
The heat release rate (blue, kW) and four thermocouple temperatures (°C). The same curves FDS wrote to its output files, now inside the viewer.

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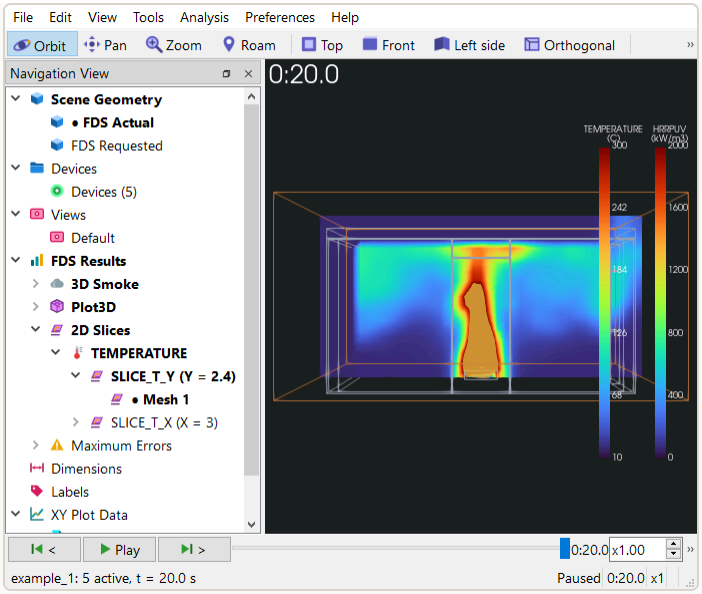
10 · Views, section planes, dimensions and labels

MENU View > Top / Front / Side / Reset Camera · View > Set Section Plane… · Tools > Add Dimension… / Add Label… · View > New View…

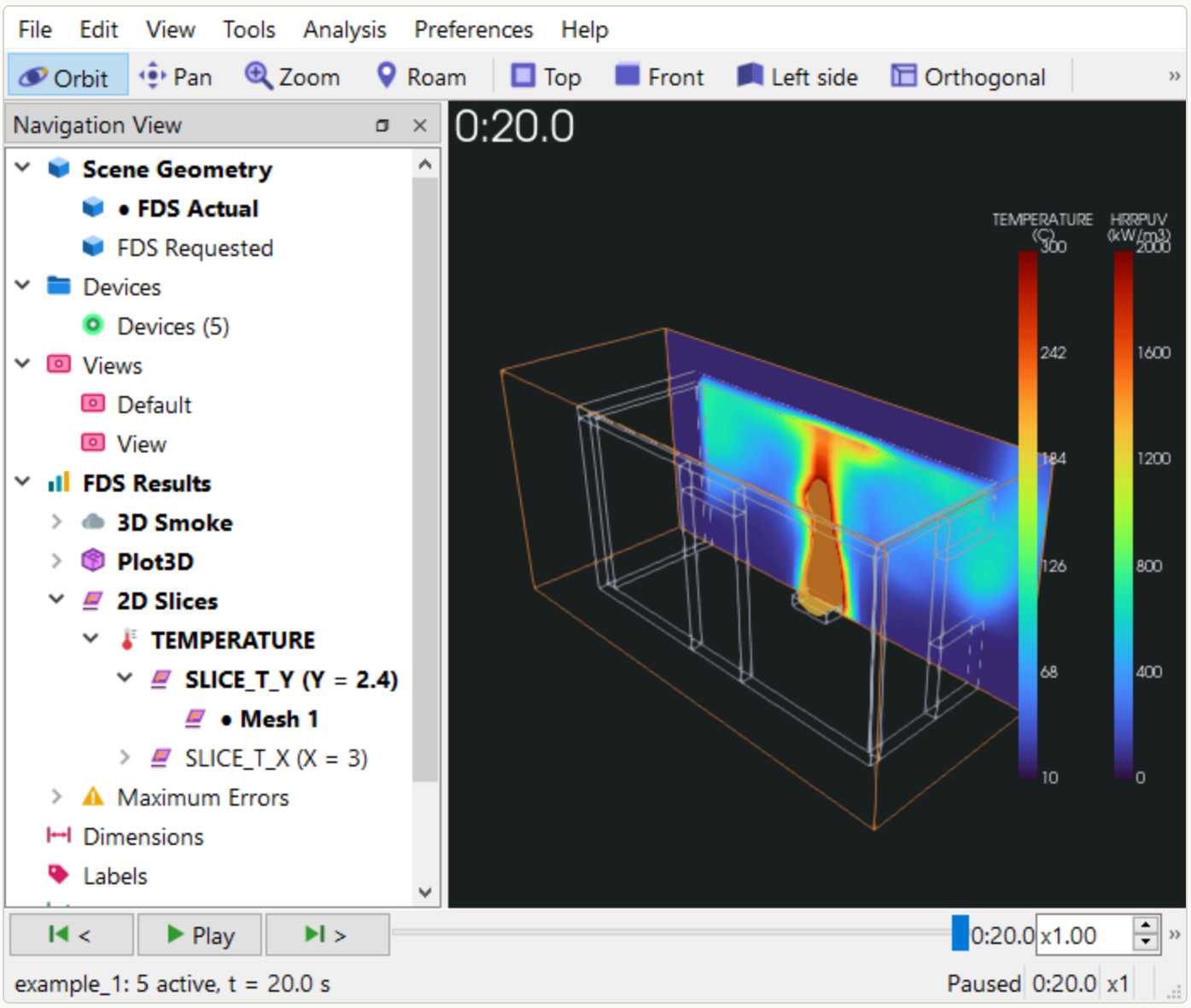
Use the toolbar or the View menu to jump to the standard directions. A **section plane** cuts the model away on one side, like slicing a cake, so you can see inside without moving the camera. A **dimension** draws a measured line between two points; a **label** pins a text to a point. **New View** stores the current camera position under a name, so you can come back to it, or use it for screenshots and movies.



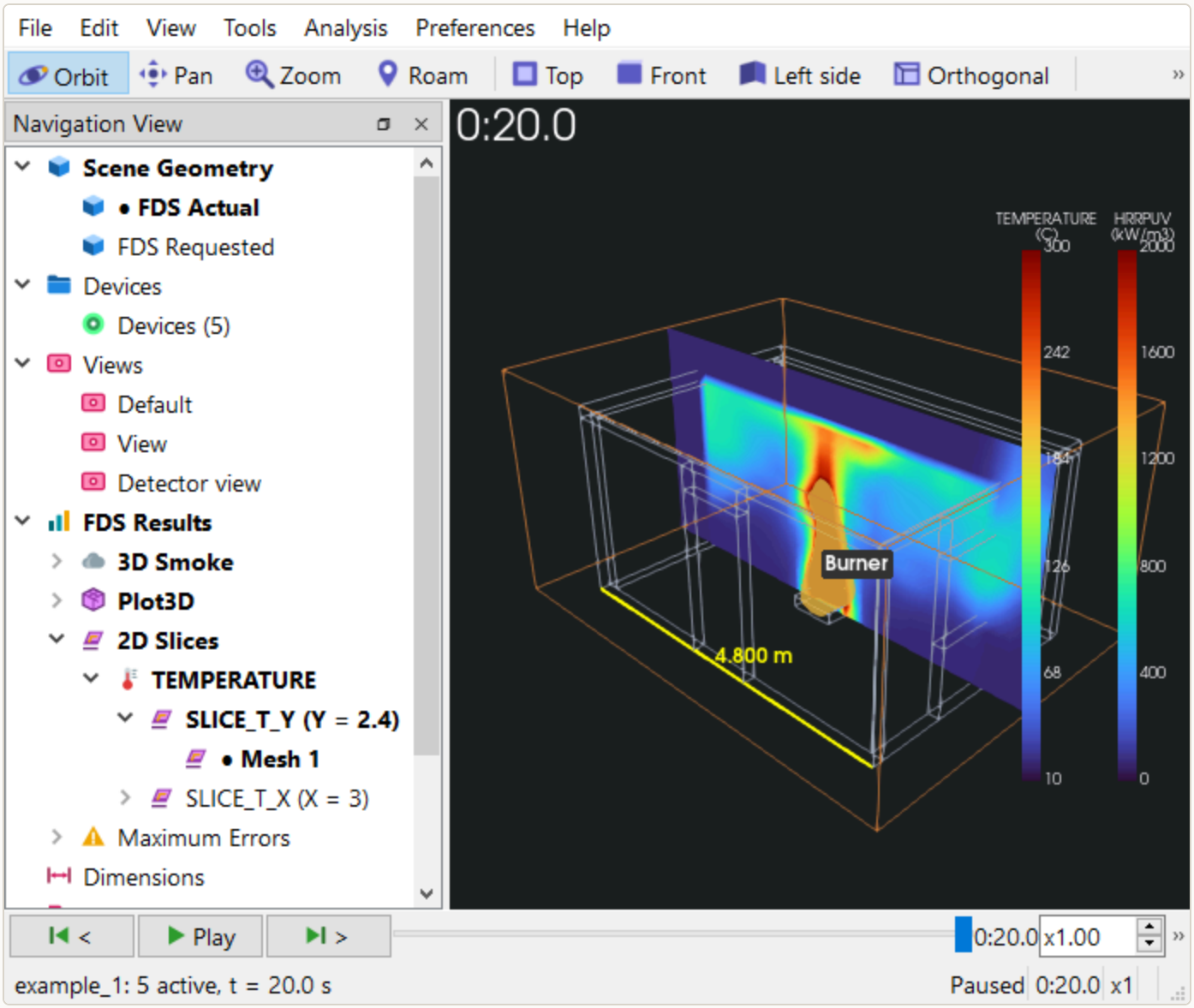
Top view.



Front view.



A section plane at y = 2.4 m: the far half of the room is removed and the plume is seen from the side.



A dimension (4.800 m along the south wall) and a saved view named “Detector view”, both listed in the tree.

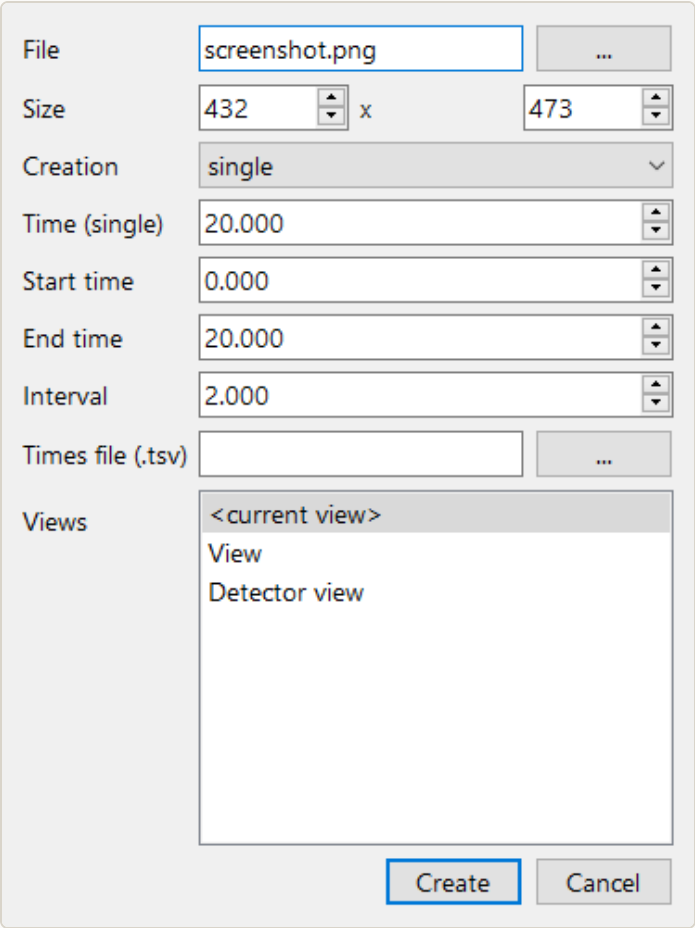
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11 · Screenshots and the animation

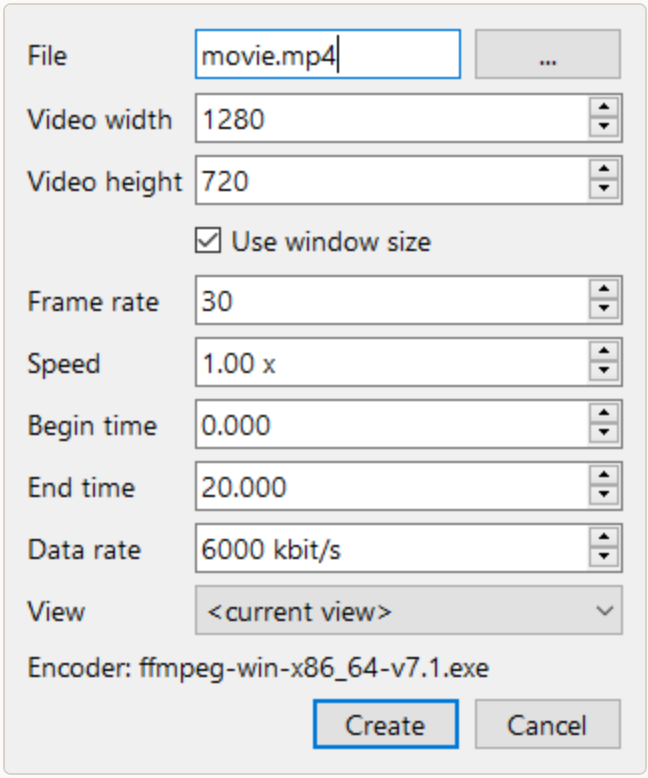
MENU File > Save Screenshot... (Ctrl+P) · File > Save Movie... · File > Save 360 Movie...

Screenshot. The dialog asks for a file name, the picture size, and how many pictures to make: a **single** picture at one time, **multiple** pictures between a start and end time at a fixed interval, or pictures at the times listed in a text file. You can also choose which saved views to photograph.

Movie. The dialog asks for the file name, size, frame rate, speed (2× means two seconds of fire per second of film), begin and end time, data rate and which view to use. The viewer draws each frame itself and joins them into an MP4 file. For this case the run was made into a film of **101 frames** at 10 frames per second and 2× speed (about 10 seconds long, 960 × 540 pixels). It is saved as `gui_viewer/example_1.mp4` . **Save 360 Movie** does the same for a 360° picture that can be looked around in a VR or 360° player.

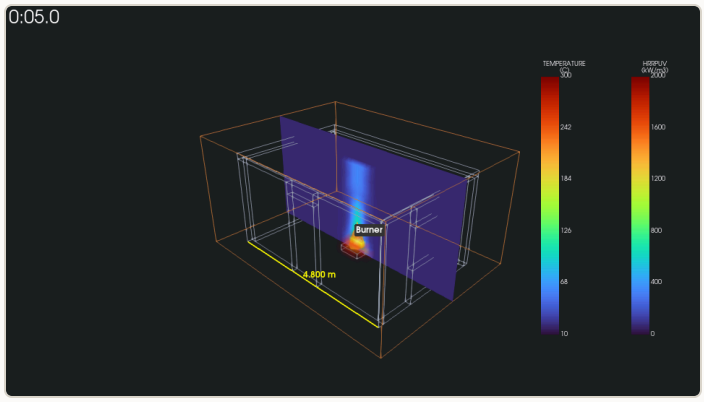


Save Screenshot dialog.

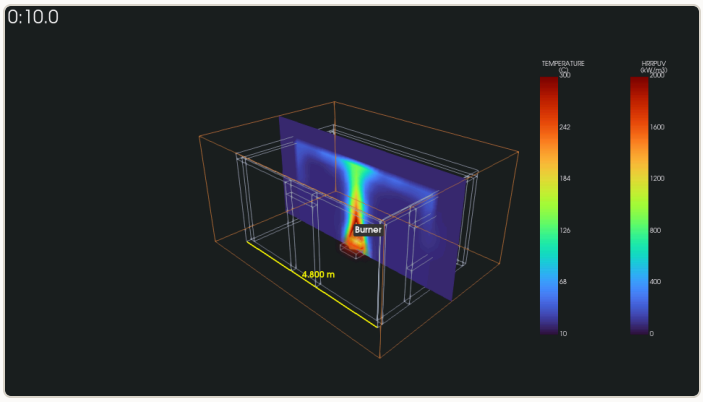


Save Movie dialog. The line at the bottom shows which video encoder was found.

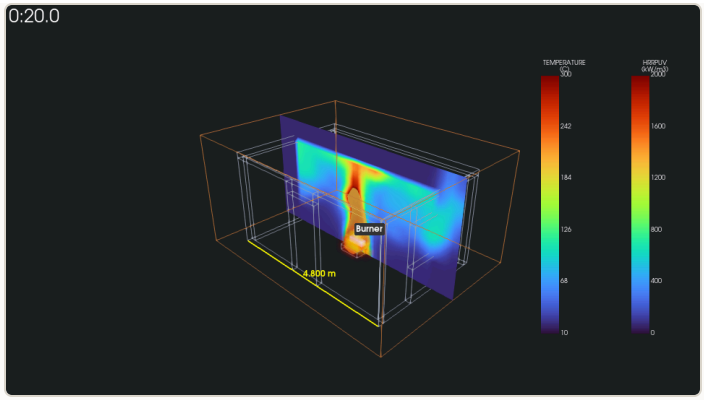
Three frames from the film (the temperature slice on the plane y = 2.4 m together with the flame, the burner label and the room dimension):



5 s



10 s



20 s

The same can be done without opening the window, for example for a batch of cases: `python -m pyfds.gui.results -movie -mvname film.mp4 example_1.smv` (add `-mv360` for a 360° film, or `-screenshot` for pictures).

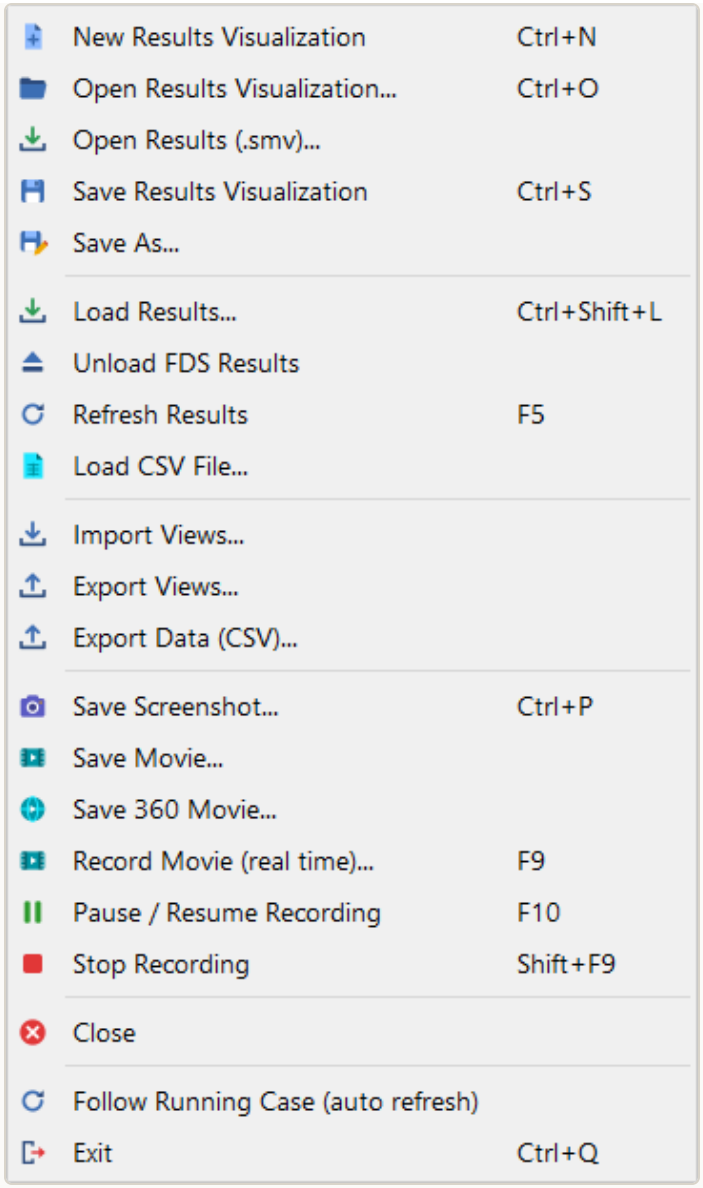
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12 · Record what you see, in real time

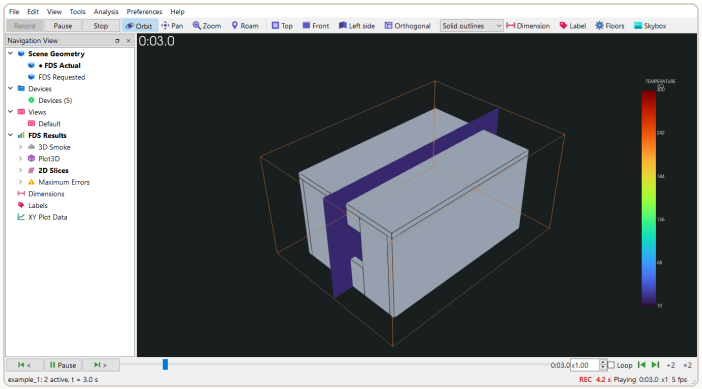
`MENU` File > Record Movie (real time)... (F9) · Pause / Resume Recording (F10) · Stop Recording (Shift+F9)

Save Movie draws every frame again, offline. **Record** is different: it films the 3D view as you use it, like a screen recorder for the scene only. Set up the picture, press **Record** (or F9) and choose the file name. Then play the run, orbit, pan, zoom, switch objects on and off: everything that happens in the 3D view goes into the film, including the clock in the corner.

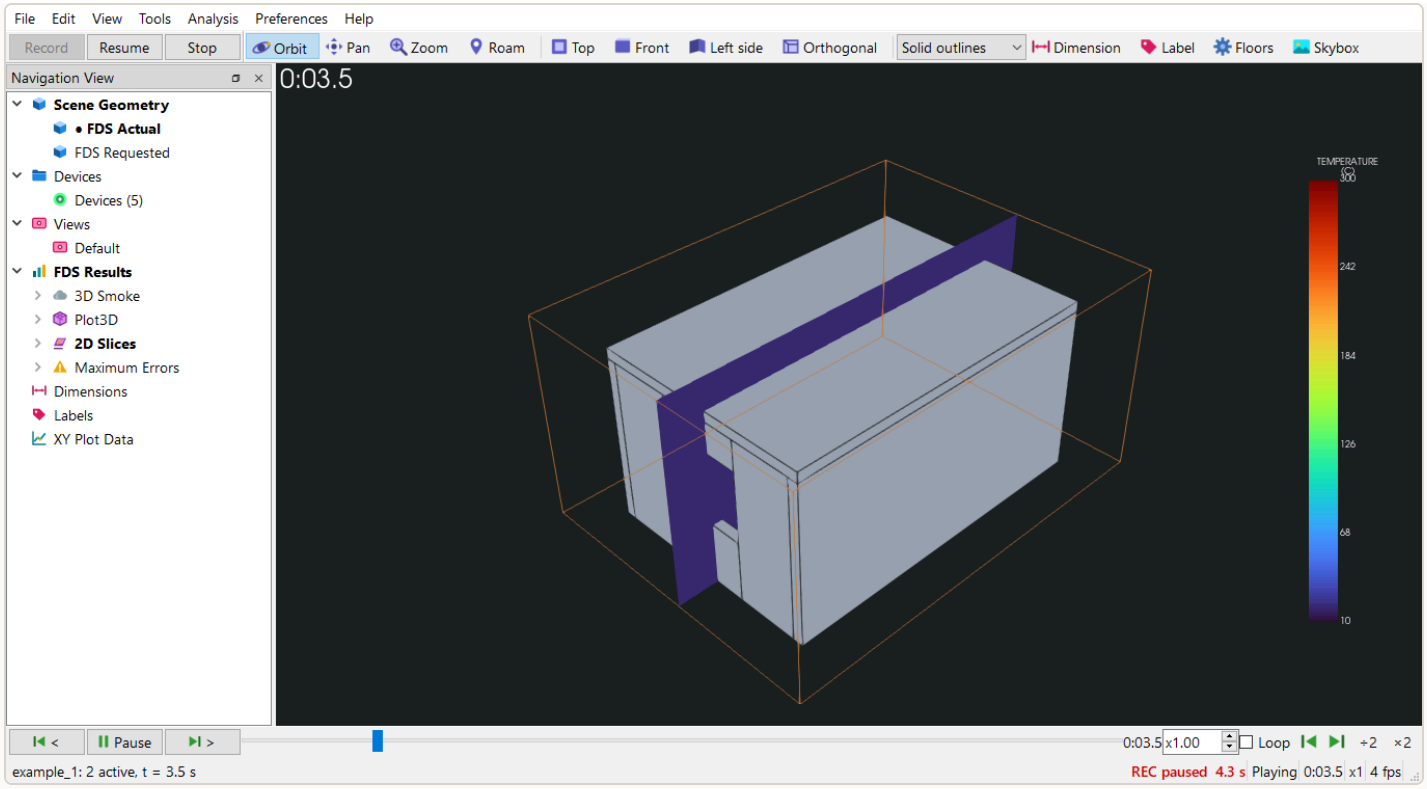
The **Record**, **Pause** and **Stop** buttons sit in the bar above the 3D view, and the status bar shows a red **REC** with the seconds recorded so far. **Pause** (F10) stops the film without ending it; the time you spend paused is not in the film, so you can rearrange the scene and carry on. **Stop** (Shift+F9) closes the file.



The recording commands in the File menu.



Recording while the run plays: the buttons above the scene, and REC with the running time at the bottom right.



Paused: the button now reads **Resume** and the status bar shows “REC paused”.

- **The film runs in real time.** It is written at 15 frames per second. If the computer draws fewer pictures than that, the last picture is repeated, so one second of film is one second on the clock and the film never runs fast. Slow drawing shows up as a less smooth film, not as a wrong speed.
- **Size.** The film has the size of the 3D view when you press Record. Do not resize the window while recording; the film keeps its first size and crops or pads.
- **Format.** MP4 (H.264) through the video encoder that comes with PyFDS. If no encoder is found, numbered PNG pictures are kept in a folder beside the chosen name instead.
- **Closing the window** while recording ends the film cleanly; what was recorded is kept.

Example

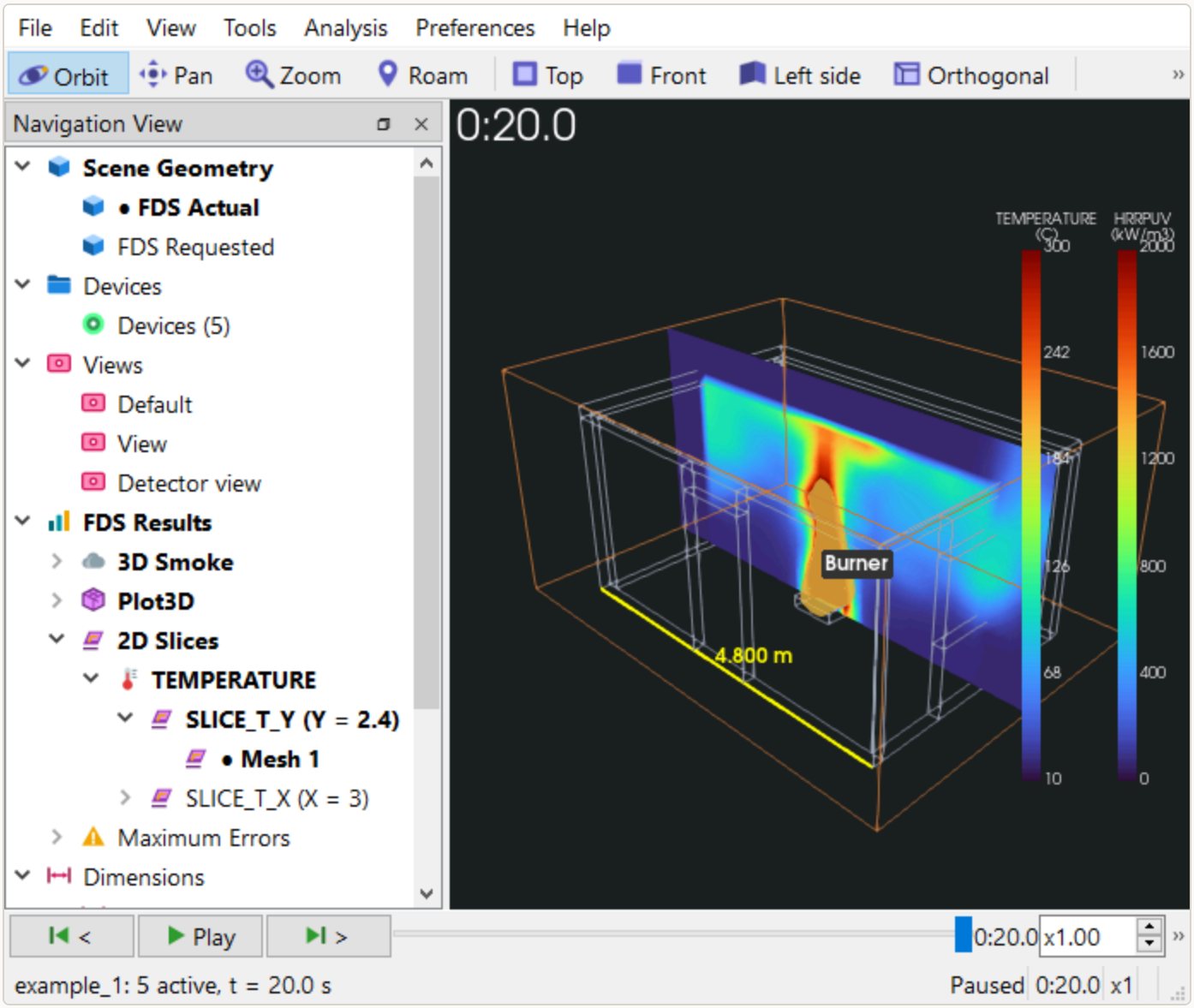
On this case a 5.9 s recording of the slice playing at ×1 gave 86 frames. It is saved as `gui_viewer/example_1_live.mp4`. Use **Save Movie** when you want an exact film of the whole run at a chosen frame rate and size, and **Record** when you want to show a walk-through of the scene.

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13 · Save the visualization

`MENU` File > Save As... → `example_1.pyfdsv`

Everything set up in the viewer (which items are on, the colour bars, views, dimensions, labels, section planes, loaded CSV files, preferences) can be saved in one small text file with the extension **.pyfdsv**. Opening it later with File > Open Results Visualization brings back the same picture. The file refers to the run's folder; it does not copy the data.



Saved. The window title shows the file name; an asterisk appears there when there are unsaved changes.

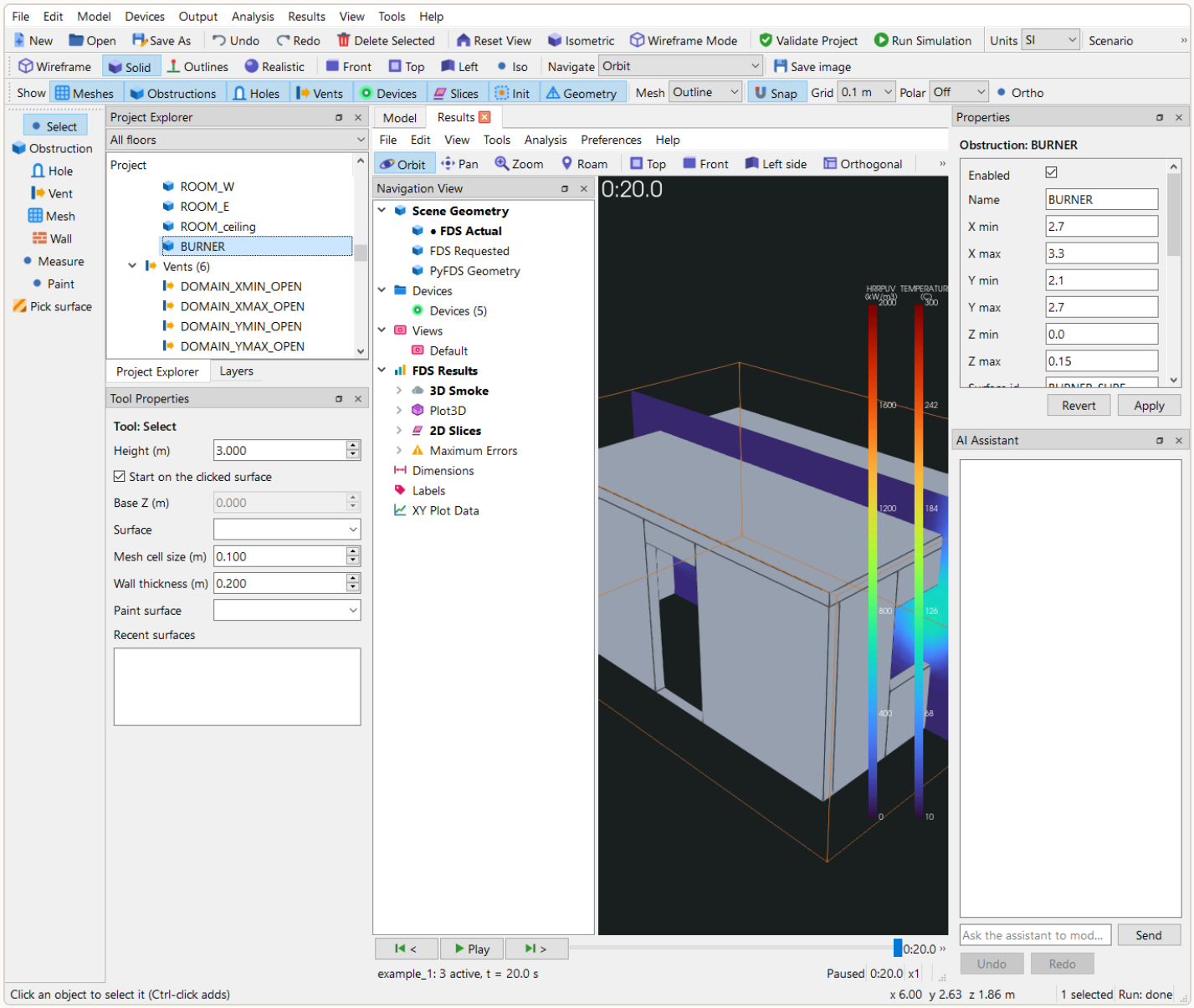
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14 · The viewer inside the main window

PANEL Messages > Results > Open in Results Viewer · or Run Monitor > Open Results Viewer

The viewer does not have to be a second window. The **Results** tab of the bottom **Messages** panel lists the result sets (`.smv`) of the last run, and opening one shows the 3D Results Viewer as a tab

named **Results** beside the **Model** tab. The model, the Project Explorer and the Properties panel stay around it, so you can click **Model**, change something, and come back. Everything described in this manual works the same inside the tab. Close the tab with its x when you are done; it can be opened again at any time.



The finished run in the Results tab of the main window.

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15 · If playback is slow or seems stuck

- **Smoke or flame on an older graphics card.** The viewer measures the first picture. If it needs more than a third of a second, it switches by itself to the **compatibility smoke renderer**

(software, about 15 times faster on the laptop used for this manual) and says so in the status bar. You can also choose it yourself: Preferences > Rendering... > **Max Compatibility**.

- **Too much on at once.** Switch off what you do not need to see (double-click it in the tree). A slice is cheap; a 3D smoke volume costs the most.
- **Smaller window, faster picture.** Drawing time grows with the number of pixels. Make the viewer window smaller for playback, or set the picture size separately when you save a movie.
- **Speed.** Raise the speed box to ×2 or ×4. Frames are skipped, not slowed down, and the clock stays honest.
- **The window seems frozen after Play.** It is busy drawing one frame: wait a second, press Pause, and step with **>**. If the frame rate in the status bar stays below 1 fps, apply the compatibility renderer above.

For the record

Before this was fixed the same scene took about 12 seconds per frame on that laptop: every smoke object made the graphics card draw the whole picture again (six times per frame), and the time stamps of all 655 frames were read from disk again for each frame. Now there is one picture per frame, the stamps are read once, and the software renderer is chosen automatically when the card is slow.